

Beyond the Dense Layer

So far, all the layers we have used have been **Dense layers**.

- **Dense Layer (Recap):** In a dense layer, every neuron is **fully connected**, meaning it receives input from *all* the activations in the previous layer.

While Dense layers are powerful, some problems are better solved by more specialized layer types.

The Convolutional Layer

A very common and powerful alternative is the **convolutional layer**.

- **Core Idea:** In a convolutional layer, each neuron is **not** connected to all inputs. Instead, it is only connected to a small, localized **region** (or "window") of the input.
- A network that uses these layers is called a **Convolutional Neural Network (CNN)**.

Example 1: 2D Image (Handwritten Digit)

- **Input:** A large grid of pixels.
- **Convolutional Layer:**
 - Neuron 1 might only look at a small 10x10 pixel square in the **top-left corner**.
 - Neuron 2 might only look at a 10x10 pixel square in the **top-right corner**.
 - ...and so on. Each neuron focuses on a different small patch of the image.

Benefits of Convolutional Layers

1. **Speeds up computation:** Because each neuron has far fewer connections, the math is much faster.
 2. **Needs less training data:** The network makes assumptions about the data (that local patterns matter), which reduces the amount of data it needs to learn.
 3. **Less prone to overfitting:** This is a key advantage, which will be discussed more next week.
-

Example 2: 1D Signal (EKG Classification)

Convolutional layers aren't just for 2D images; they work well for any data with a spatial or sequential structure, like 1D time-series data.

- **Problem:** Classify a patient's EKG (heartbeat) signal.

- **Input:** A 1D vector of 100 signal measurements.
 - **Layer 1 (Convolutional):**
 - Neuron 1 looks at inputs 1–20.
 - Neuron 2 looks at inputs 11–30.
 - Neuron 3 looks at inputs 21–40.
 - ...and so on. Each neuron looks at a small, overlapping "window" of the EKG signal.
 - **Layer 2 (Convolutional):**
 - This layer can *also* be convolutional. It takes the activations from Layer 1.
 - Neuron 1 (in Layer 2) might look at activations 1–5 from Layer 1.
 - Neuron 2 (in Layer 2) might look at activations 3–7 from Layer 1.
 - **Output Layer (Dense):**
 - Finally, a Dense (sigmoid) layer can be added at the end. This layer *would* look at all the activations from the last convolutional layer to make the final binary classification (e.g., "heart disease" or "no heart disease").
-

Conclusion

- You **do not** need to know how to implement convolutional layers for this course.
- The key takeaway is that the Dense layer is just one of many "building blocks" for a neural network.
- Much of modern AI research (e.g., **Transformers**, LSTMs, etc.) involves inventing new, specialized layer types to solve specific kinds of problems.